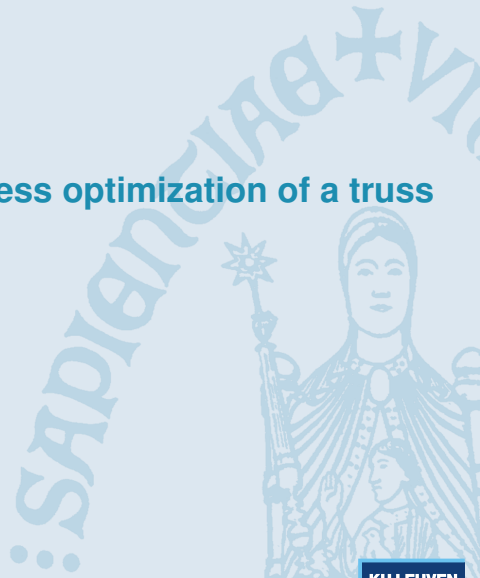


Chapter 5 – Sizing stiffness optimization of a truss



5.1 The simultaneous formulation of the problem

- Minimize compliance C subject to a volume constraint $V \leq V_{\max}$:

$$\min_{\mathbf{x}, \mathbf{u}} : C = \mathbf{F}^T \mathbf{u}$$

$$\text{s.t.} : \mathbf{K}(\mathbf{x}) \mathbf{u} = \mathbf{F}$$

$$\sum_{j=1}^n l_j x_j \leq V_{\max}$$

$$\mathbf{x} \in \mathcal{X} = \{\mathbf{x} \in \mathbb{R}^n : x_j^{\min} \leq x_j \leq x_j^{\max}, j = 1, \dots, n\}$$

$$\text{where } \mathbf{K}(\mathbf{x}) = \sum_{j=1}^n x_j \mathbf{K}_j^0 = \sum_{j=1}^n x_j \mathbf{C}_j^T \mathbf{k}_j^0 \mathbf{C}_j$$

where $\mathbf{k}_j^0$ is the element stiffness matrix for element j with unit section area.

- Alternative expressions for the compliance:

$$C = \mathbf{F}^T \mathbf{u} = \mathbf{u}^T \mathbf{K} \mathbf{u} = \sum_{j=1}^n x_j \mathbf{u}^T \mathbf{K}_j^0 \mathbf{u} = \sum_{j=1}^n x_j \mathbf{u}_j^T \mathbf{k}_j^0 \mathbf{u}_j$$

5.1 The simultaneous formulation of the problem

- The simultaneous formulation of the problems is non-convex:
 - The equilibrium equations $\mathbf{h}(\mathbf{x}, \mathbf{u}) = \mathbf{K}(\mathbf{x})\mathbf{u} - \mathbf{F} = \mathbf{0}$ can be written as 2 sets of inequalities $\mathbf{h}(\mathbf{x}, \mathbf{u}) \leq \mathbf{0}$ and $-\mathbf{h}(\mathbf{x}, \mathbf{u}) \leq \mathbf{0}$.
 - In order for the feasible set to be convex, both $\mathbf{h}$ and $-\mathbf{h}$ must be convex.
 - This means that $\mathbf{h}$ must be affine in $\mathbf{x}$ and $\mathbf{u}$, i.e. it may be written as $\mathbf{h}(\mathbf{x}, \mathbf{u}) = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{c}$ for some constant matrices $\mathbf{A}$ and $\mathbf{B}$ and a constant vector $\mathbf{c}$.
 - This is not possible; as a consequence, the problem is not convex.

5.2 The nested formulation of the problem

- Minimize compliance C subject to a volume constraint $V \leq V_{\max}$:

$$\min_{\mathbf{x}} : C = \mathbf{F}^T \mathbf{u}(\mathbf{x})$$

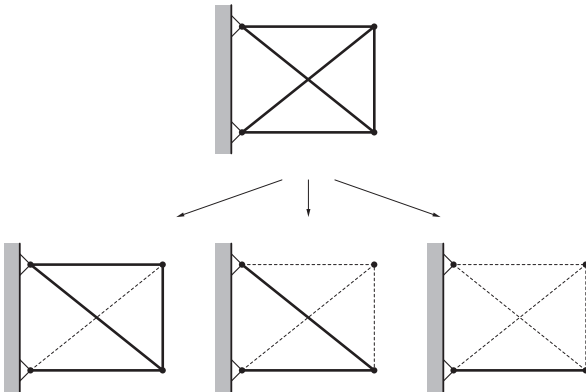
$$\text{s.t. : } \sum_{j=1}^n l_j x_j \leq V_{\max}$$

$$\mathbf{x} \in \mathcal{X} = \{\mathbf{x} \in \mathbb{R}^n : x_j^{\min} \leq x_j \leq x_j^{\max}, j = 1, \dots, n\}$$

$$\text{where } \mathbf{u}(\mathbf{x}) = \mathbf{K}^{-1}(\mathbf{x})\mathbf{F}(\mathbf{x})$$

5.2 The nested formulation of the problem

- The nested formulation is only applicable if the stiffness matrix $\mathbf{K}(\mathbf{x})$ does not become singular for the optimal truss.



- Solution: use a very small lower bound $x_j^{\min} = \varepsilon > 0$.

5.2 The nested formulation of the problem

- Gradient of the compliance C (assuming that $\mathbf{F}$ does not depend on x_j):

$$\frac{\partial C}{\partial x_j} = \mathbf{F}^T \frac{\partial \mathbf{u}}{\partial x_j}$$

The term $\frac{\partial \mathbf{u}}{\partial x_j}$ is obtained as follows:

$$\mathbf{K}\mathbf{u} = \mathbf{F} \Rightarrow \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} + \mathbf{K} \frac{\partial \mathbf{u}}{\partial x_j} = \mathbf{0} \Leftrightarrow \frac{\partial \mathbf{u}}{\partial x_j} = -\mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u}$$

This equation is introduced in the expression for $\frac{\partial C}{\partial x_j}$:

$$\frac{\partial C}{\partial x_j} = -\mathbf{F}^T \mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} = -\mathbf{u}^T \frac{\partial \mathbf{K}}{\partial x_j} \mathbf{u} = -\mathbf{u}^T \mathbf{K}_j^0 \mathbf{u} = -\mathbf{u}_j^T \mathbf{k}_j^0 \mathbf{u}_j$$

- Hessian of the compliance C :

$$\frac{\partial^2 C}{\partial x_i \partial x_j} = 2\mathbf{u}(\mathbf{x})^T \mathbf{K}_i^0 \mathbf{K}(\mathbf{x})^{-1} \mathbf{K}_j^0 \mathbf{u}(\mathbf{x})$$

5.2.1 Convexity of the nested problem

- In order to verify that the Hessian $\nabla^2 C(\mathbf{x})$ is positive semidefinite, consider:

$$\begin{aligned}\mathbf{y}^T \nabla^2 C(\mathbf{x}) \mathbf{y} &= \sum_{j=1}^n \sum_{i=1}^n \frac{\partial^2 C}{\partial x_i \partial x_j} y_i y_j \\ &= 2\mathbf{u}(\mathbf{x})^T \left[\sum_{i=1}^n \mathbf{K}_i^0 y_i \mathbf{K}(\mathbf{x})^{-1} \sum_{j=1}^n \mathbf{K}_j^0 y_j \right] \mathbf{u}(\mathbf{x})\end{aligned}$$

- Introduce the symmetric matrix $\mathbf{Y} = \sum_{i=1}^n \mathbf{K}_i^0 y_i$:

$$\begin{aligned}\mathbf{y}^T \nabla^2 C(\mathbf{x}) \mathbf{y} &= 2\mathbf{u}(\mathbf{x})^T [\mathbf{Y} \mathbf{K}(\mathbf{x})^{-1} \mathbf{Y}] \mathbf{u}(\mathbf{x}) \\ &= 2 [\mathbf{Y} \mathbf{u}(\mathbf{x})]^T \mathbf{K}(\mathbf{x})^{-1} [\mathbf{Y} \mathbf{u}(\mathbf{x})] \geq 0\end{aligned}$$

where the last inequality holds because $\mathbf{K}(\mathbf{x})^{-1}$ is positive definite since $\mathbf{K}(\mathbf{x})$ is positive definite.

- Since the Hessian $\nabla^2 C(\mathbf{x})$ is positive semidefinite, the compliance is convex. As the constraint function $\sum_{j=1}^n l_j x_j \leq V_{\max}$ is also convex, the nested problem is a convex problem.

5.2.2 Fully stressed designs

- Strain energy and external work for a single bar:

→ Strain energy U_j of a single bar j :

$$U_j = \frac{1}{2} \int \sigma_j \epsilon_j dV_j = \frac{1}{2} \sigma_j \epsilon_j x_j l_j = \frac{1}{2} \frac{\sigma_j^2}{E_j} x_j l_j$$

→ Work W_j done by the external forces $\mathbf{f}_j$ acting on bar j :

$$W_j = \frac{1}{2} \mathbf{f}_j^T \mathbf{u}_j = \frac{1}{2} \mathbf{u}_j^T \mathbf{k}_j \mathbf{u}_j = \frac{1}{2} x_j \mathbf{u}_j^T \mathbf{k}_j^0 \mathbf{u}_j$$

→ Conservation of energy $U_j = W_j$:

$$\mathbf{u}_j^T \mathbf{k}_j^0 \mathbf{u}_j = \frac{\sigma_j^2}{E} l_j$$

- Gradient of the compliance C in terms of σ_j :

$$\frac{\partial C}{\partial x_j} = -\mathbf{u}_j^T \mathbf{k}_j^0 \mathbf{u}_j = -\frac{\sigma_j^2}{E} l_j$$

5.2.2 Fully stressed designs

- Lagrangian of the nested problem:

$$\mathcal{L}(\mathbf{x}, \lambda) = C(\mathbf{x}) + \lambda \left(\sum_{i=1}^n x_i l_i - V_{\max} \right)$$

- Differentiation of the Lagrangian with respect to x_j :

$$\frac{\partial \mathcal{L}(\mathbf{x}, \lambda)}{\partial x_j} = -\frac{\sigma_j^2}{E} l_j + \lambda l_j$$

- The KKT conditions give the following KKT point $(\mathbf{x}^*, \lambda^*)$:

$$\begin{aligned} \sigma_j^2 &\leq \frac{\lambda^*}{E}, & \text{if } x_j^* &= x_j^{\min} \\ \sigma_j^2 &= \frac{\lambda^*}{E}, & \text{if } x_j^{\min} < x_j^* < x_j^{\max} \\ \sigma_j^2 &\geq \frac{\lambda^*}{E}, & \text{if } x_j^* &= x_j^{\max} \end{aligned}$$

5.2.3 Minimization of volume under compliance constraint

- Minimization of compliance under volume constraint (A):

$$\begin{aligned} \min_{\mathbf{x}} : \quad & C = \mathbf{F}^T \mathbf{u}(\mathbf{x}) \\ \text{s.t.} : \quad & \sum_{j=1}^n l_j x_j \leq V_{\max} \\ & \mathbf{x} \in \mathcal{X} = \{\mathbf{x} \in \mathbb{R}^n : x_j^{\min} \leq x_j \leq x_j^{\max}, j = 1, \dots, n\} \end{aligned}$$

- Minimization of volume under compliance constraint (B):

$$\begin{aligned} \min_{\mathbf{x}} : \quad & V = \sum_{j=1}^n l_j x_j \\ \text{s.t.} : \quad & C = \mathbf{F}^T \mathbf{u}(\mathbf{x}) \leq C_{\max} \\ & \mathbf{x} \in \mathcal{X} = \{\mathbf{x} \in \mathbb{R}^n : x_j^{\min} \leq x_j \leq x_j^{\max}, j = 1, \dots, n\} \end{aligned}$$

5.2.3 Minimization of volume under compliance constraint

- If the box constraints are inactive, (A) and (B) yield the same solution except for a scaling factor:

$$\mathbf{x}_B^* = \frac{C_A^*}{C_{\max}} \mathbf{x}_A^*$$
$$\mathbf{x}_A^* = \frac{V_{\max}}{V_B^*} \mathbf{x}_B^*$$

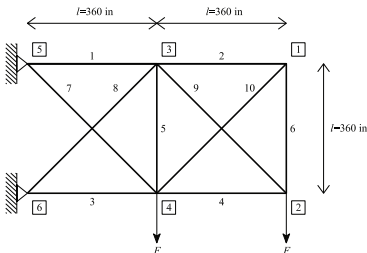
Proof: see textbook. The proof is based on the following lemma:

Let $\mathbf{K}(\mathbf{x})\mathbf{u} = \mathbf{F}$ and $\mathbf{K}(\mathbf{x}^*)\mathbf{u}^* = \mathbf{F}$, where $\mathbf{x}^* = \alpha\mathbf{x}$.

Then $\mathbf{u}^* = \frac{1}{\alpha}\mathbf{u}$.

5.3 Numerical solution of the nested problem using MMA

Minimization of the compliance of a ten-bar truss subject to a volume constraint.



$$E = 10^7 \text{ psi} = 68.94 \times 10^9 \text{ N/m}^2$$

$$F = 100 \text{ kips} = 444822 \text{ N}$$

$$V_{\max} = 0.4 \text{ m}^3$$

$$x_{\min} = 10^{-4} \text{ m}^2$$

$$x_{\max} = 0.1 \text{ m}^2$$

Solution in MATLAB using MMA:

$$\min_{\mathbf{x}} : f_0(\mathbf{x})$$

$$\text{s.t.} : f_1(\mathbf{x}) \leq 0$$

$$x_j^{\min} \leq x_j \leq x_j^{\max}$$

where:

$$f_0(\mathbf{x}) = 10^{-4} C$$

$$\frac{\partial f_0}{\partial x_j} = 10^{-4} \frac{\partial C}{\partial x_j}$$

$$f_1(\mathbf{x}) = \frac{V}{V_{\max}} - 1$$

$$\frac{\partial f_1}{\partial x_j} = \frac{1}{V_{\max}} \frac{\partial V}{\partial x_j}$$

The objective function is scaled to comply with the practical guidelines for MMA; see the accompanying PDF-file on smoptit.se.

5.3 Numerical solution of the nested problem using MMA

Minimization of the compliance of a ten-bar truss subject to a volume constraint.

- Minimize the compliance of the ten-bar truss using MMA (ex1a.m).
- Analyze the convergence history of the optimization algorithm (ex1b.m).
- Remove the load at node 4 and give a physical interpretation of the resulting design (ex1c.m).
- Reintroduce the load at node 4 and verify whether the optimized design is fully stressed (ex1d.m). Use the Matlab functions `elemforces` and `plotstress`.
- Modify the volume constraint such that the stress becomes equal to 160 N/mm^2 (ex1e.m).
- Transform the problem into a minimum volume problem with $C_{\max} = 10^4 \text{ Nm}$ and verify to which extent the statements made on slide 89 hold in this case (ex1f.m). Here you need to define the initial design in a different way – you can use the following value:

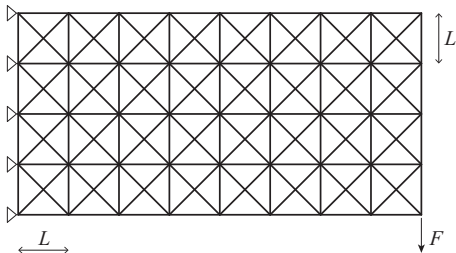
$$x_j = \frac{0.4}{\sum_{e=1}^N L_e}$$

Compare the two designs as follows: ex1a; xA=x; ex1f; xB=x; xA./xB

- Go back to the original minimum compliance problem and put $x_{\max} = 0.005 \text{ m}^2$ (ex1g.m). What is the influence on the optimized design and its performance?

5.3 Numerical solution of the nested problem using (GC)MMA

Minimization of the compliance of a 140-bar truss subject to a volume constraint.



$$L = 1 \text{ unit}$$

$$E = 1 \text{ unit}$$

$$F = 1 \text{ unit}$$

$$V_{\max} = 166.5097 \text{ units}$$

$$x_{\min} = 0.01 \text{ units}$$

$$x_{\max} = 20 \text{ units}$$

- Minimize the compliance of this 140-bar truss using MMA (ex2a).
- Minimize the compliance of this 140-bar truss using GCMMA (ex2b).
- Minimize the compliance of this 140-bar truss using GCMMA, but only compute the sensitivities when needed – in the outer iterations, not in the inner iterations (ex2c).